

CLIENT SYSTEM ARCHITECTURE & REQUIREMENTS (SIMPLIFIED)

Overview

The client is designing a complete parts management and procurement system that connects:

- Fleet machines
- Parts database
- OEM comparison worksheet
- Procurement workflow
- Quotation system
- Warehouse operations

The system will later integrate with ERP systems and the company's internal AI platform.

1. STANDARDIZED FLEET DATABASE

The Fleet Database stores information about every machine owned by customers.

This database is the foundation of the system because it allows software to identify which parts belong to each machine.

Required Fields:

- Machine Type
- Machinery Make
- Year
- Model
- VIN / Serial Number
- Engine Serial Number (only for trucks)
- Engine Arrangement Number (auto-filled by system)
- Unit Number

Example Record:

Machine Type: Motor Grader

Make: Caterpillar

Year: 2009

Model: 140M

VIN/SN: B9M00779

Engine Serial: KHX16274

Arrangement#: C16274

Unit#: _C16274

Unit Number Logic:

If the customer already has a unit number → use it.

If not → the system should generate a unit number automatically.

2. OEM PRICE COMPARISON WORKSHEET

The OEM Comparison Worksheet is a standardized Excel workbook used by multiple teams to collect and compare parts data.

Users:

- Parts technicians
- Supply chain teams
- VIP fleet operations
- AI system

System Process:

1. Customer fleet list uploaded
2. System identifies the machine
3. System retrieves correct OEM parts
4. OEM worksheet automatically populates
5. OEM MSRP pricing is displayed

Primary Uses:

- Customer quotations
- Procurement orders (RFQ → PO)
- OEM price comparison
- Supplier sourcing

Example Data Structure:

Category: Filters

Product Type: MTP-Track

Part Number: 106-1605

Description: Pads

Unit Price: XXX

Quantity: XXX

3. PROCUREMENT AND SYSTEM INTEGRATION

3A. Warehouse Procurement System

The warehouse development team will use the OEM worksheet to generate procurement orders.

Required Fields:

- Unit-Grp (Category)
- Default Unit (Product Type)
- Description
- Part Number
- Unit Price
- Quantity

Example:

Category: Filters

Product Type: MTP-Track

Part Number: 106-1605

Description: Pads

Price: XXX

Quantity: XXX

3B. Sandbox Field Service System

Sandbox is an automated workflow system that generates quotations.

Minimal data is uploaded and the system automatically fills the rest.

Required Upload Fields:

- Part Number
- Category
- Product Type
- Description
- Quantity
- Discount Per Item

Example:

Part Number: 106-1605

Category: Undercarriage

Product Type: MTP-Track

Description: Pads

Quantity: XXX

Discount: 50%

AUTOMATIC QUOTATION GENERATION

Internal View (Company Only)

Fields:

- Internal SKU
- Category
- Product Type
- Customer Selling Price
- Quantity
- Tier
- Manufacturer

Example:

Internal SKU: U-MTP-1605

Category: Undercarriage

Product Type: Track

Description: Pads

Customer Quotation View

Customer quotations hide internal data such as:

- Manufacturer selection
- Tier levels
- Internal system logic

Customer View Fields:

- Internal SKU
- Category
- Product Type
- Customer Selling Price
- Quantity

INTERNAL SKU LOGIC

Customers should not see OEM part numbers.
Instead, the system generates Internal SKUs.

Example:

OEM Part Number: 106-1605

Internal SKU: U-MTP-1605

WHY INTERNAL SKU IS IMPORTANT

Warehouse Operations

- Inventory tracking
- Stock movement
- Bin location management

Accounting

- Quotations
- Invoice tracking
- Cost and margin analysis

Software Systems

- Database relationships
- System logic
- Integration between procurement, CRM, and inventory

DATABASE STRUCTURE RULE

Primary Key: Internal SKU

OEM part numbers remain reference fields.

Example Table:

Internal SKU: U-MTP-1605

OEM Part Number: 106-1605

Category: Undercarriage

Product Type: Track

OVERALL SYSTEM FLOW

Customer Fleet List



Fleet Database



Machine Selected



OEM Parts Retrieved



OEM Price Comparison Worksheet



Sandbox Upload

↓

Automatic Quotation Generation

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Customer Approval

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Procurement Order

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Warehouse Processing

YOUR ROLE IN THIS PROJECT

Your development work focuses on:

1. Improving the OEM Comparison Excel workbook.
2. Structuring parts and fleet data correctly.
3. Preparing the data structure for integration with procurement systems and future ERP systems.

The Excel workbook will act as the main bridge between fleet data, the parts database, and procurement workflows.